

Harsh Bordekar

AI Research Engineer - ML Engineering

Deep Learning Infrastructure, HPC-Scale Simulation Pipelines & Custom Model Architectures

PhD Candidate, DLR & TU Braunschweig — Aerospace Composites & Structural Simulation

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🌐 [Harsh's LinkedIn](#)

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🐙 github.com/harshbordekar-stack

🏠 Permanent Resident - Germany

PROFILE

Simulation and machine-learning research engineer with 7+ years building physics-based models and automation pipelines for aerospace structures, including 3+ years working hands-on across the applied deep-learning stack (PyTorch, TensorFlow, scikit-learn), building custom model architectures end-to-end rather than only calling pretrained ones, from data pipeline through training to a deployed, tool-calling interface. PhD work runs large-scale simulation campaigns on HPC/SLURM clusters, generating and managing the high-fidelity, multiscale datasets a probabilistic surrogate framework is trained and evaluated against, where compute scheduling, job turnaround, and data volume are constant practical constraints rather than abstract concerns. Built a physics-based CNN surrogate wrapped in an agentic, tool-calling interface backed by a hierarchical retrieval pipeline, and a separate CT-based defect-detection model trained end-to-end and validated against ground truth for engineering sign-off. Comfortable in Python and C++, with a track record of verifying externally produced models against defined acceptance criteria and diagnosing numerical and performance issues down to their root cause. Experienced explaining simulation and ML concepts to both technical and non-technical audiences, including instructing 60+ Master's students.

Core Competencies: Deep Learning & Model Architecture Design (PyTorch, TensorFlow), Physics-Informed / Surrogate Modelling, Large-Scale Simulation Dataset Generation & Management, HPC Workload Scheduling (SLURM), High-Performance Pipelines (Python, C++), Model Verification, Debugging & Root-Cause Diagnosis, Agentic / Tool-Calling Systems, Docker, Git, Explainable AI

PROFESSIONAL EXPERIENCE

Structural Analysis / Simulation Engineer

DLR & TU Braunschweig

ML for simulation, HPC-scale data pipelines & surrogate modelling (PhD)

Feb 2021 - Present

- Built a physics-based CNN surrogate for microstructure-informed computational homogenization, wrapped in an agentic, tool-calling interface that accepts natural-language queries and microstructure images, retrieves context via a hierarchical RAG pipeline over **50** materials documents, and returns homogenized/effective material properties, an end-to-end system spanning data ingestion, model, and served interface. Retrieval/prediction faithfulness validated at **93%** against ground-truth homogenization results on a held-out set of more than **100** cases.
Tools: Python, Physics-Based CNN, Hierarchical RAG, LLM Agent Framework, Image Processing
- Ran large-scale thermal and structural simulation campaigns on HPC/SLURM infrastructure for the Callisto reusable-rocket programme's metal upper housing bay, scheduling and managing concurrent compute-heavy jobs across the cluster under production-scale turnaround constraints.
Tools: ABAQUS, SLURM (HPC), Python, Nastran, Patran
- (PhD) Developed a probabilistic, physics-based multiscale surrogate framework for leakage-onset prediction in CFRP hydrogen storage tanks, coupling microstructure-informed mesoscale damage models with stochastic spatial defect distributions to generate and propagate a large simulation dataset across ply, laminate, and structural scales, replacing repeated full-scale FEM runs with a physics-consistent surrogate.
Tools: Python, ABAQUS, FEM, HPC/SLURM, Multiscale Modelling
- Verified and technically reviewed externally and student-produced simulation models, meshes, and results against defined methods and acceptance criteria, diagnosing failure modes traced to subtle numerical, meshing, or data-quality issues and issuing documented sign-off-style findings.
Tools: ABAQUS, Nastran, Python
- Derived a certification-aligned leakage failure criterion mapping combined strain/load states to leakage onset, providing the evaluation standard against which the surrogate framework's coverage, accuracy, and failure modes are assessed.
Tools: ABAQUS, Python, FEM post-processing using C++
- Performed Monte Carlo uncertainty quantification over the multiscale simulation outputs to generate leakage-probability-vs-strain relationships, characterising prediction confidence for risk-informed design decisions.
Tools: Python (SciPy, NumPy), Monte Carlo methods, ABAQUS
- Instructed 60+ Master's students in FEM theory and ABAQUS-based CFRP modelling, covering Continuum Damage Mechanics (CDM), Progressive Damage Analysis (PDA), and Cohesive Zone Modelling (CZM), communicating complex simulation and ML workflows to a technical audience.
Tools: ABAQUS, Automation using Python

- Developed a CT-based, explainable-AI defect-detection framework, training and validating a classifier end-to-end to **91%** classification accuracy (**89%** precision / **91%** recall) on **1200** CT scans of additively manufactured titanium parts, cutting manual review time to **<0.5 sec per image**, with SHAP, Grad-CAM and LIME outputs used to support engineering sign-off. (*Published: Journal of Intelligent Manufacturing, 2025.*)
Tools: Python (scikit-learn, TensorFlow), SHAP, Grad-CAM, LIME, CT image processing
- Assessed fatigue and defect tolerance of aerospace-grade titanium components using small-defect fatigue models (Murakami, El-Haddad), coupling defect measurements with numerical simulation to translate physics-based fracture-mechanics criteria into allowable-defect acceptance limits.
Tools: Small-defect fatigue models (Murakami, El-Haddad), Python (NumPy/SciPy), FEM (ABAQUS)

- Developed an ML framework for automated detection and quantification of fiber undulations in unidirectional CFRP laminates from image data, generating spatially resolved defect metrics used directly as high-fidelity inputs for probabilistic structural simulation models.
Tools: MATLAB, Python (scikit-learn, TensorFlow), image processing, NumPy/SciPy

SKILLS SUMMARY

Deep Learning & ML Frameworks	PyTorch TensorFlow Scikit-learn Custom Model Architectures (Physics-Based CNN) Explainable AI (SHAP, Grad-CAM, LIME)
Programming Languages	Python C++ FORTRAN MATLAB
HPC & Simulation Infrastructure	HPC / SLURM (Large-Scale Job Scheduling) Large-Scale Simulation Dataset Generation FE Solvers (ABAQUS, Nastran/Patran)
Model Verification & Debugging	Verification & Validation Numerical / Performance Root-Cause Diagnosis Uncertainty Quantification (Monte Carlo)
LLMs & Agentic Systems	Retrieval-Augmented Generation (RAG) LangChain LangGraph Model Context Protocol (MCP) Tool-Calling
Data Analysis & Scientific Computing	NumPy Pandas Matplotlib Seaborn Plotly SALib SciPy
APIs & Backend Development	RESTful APIs FastAPI JSON
Version Control & Reproducibility	Git Docker Linux
Application Development	VS Code JupyterLab Google Colab Streamlit Gradio
Languages	English (C1) German (B1)

AWARDS AND EDUCATION

 2nd Prize: Best Paper Award	DLR (German Aerospace Center) 2026
 3rd Prize: Best Paper Award	DLR (German Aerospace Center) 2024
PhD Candidate, Computational Methods in Engineering	DLR & TU Braunschweig, Germany Feb 2021 – 2026 (expected)
Dissertation: Physics-Informed, Multiscale Machine Learning for CFRP Composite Structures	
M.Sc. Computational Methods in Engineering	Leibniz University Hannover, Germany April 2018 – Dec 2020
Focus: Numerical Methods, Scientific Machine Learning, Deep Learning	

SELECTED PUBLICATIONS

1. **Bordekar, H.**, et al. “Microstructure-informed Probabilistic Multiscale Modelling of CFRP via Machine-Learning-Derived Stochastic Volume Elements.” *Manuscript in preparation*, 2026.
2. **Bordekar, H.**, et al. “MicroStructAgent: A Physics-Informed Agentic AI Framework for Computational Homogenization and Quality Assurance of Fiber Composite Laminates.” *Manuscript in preparation*, 2026.
3. **Bordekar, H.**, et al. “Microstructure Characterization of Fiber Composite Laminate Using eXplainable Artificial Intelligence.” *Journal of Composite Materials*, vol. 60, no. 1, 2026, pp. 3–25. [DOI](#)
4. **Bordekar, H.**, et al. “eXplainable Artificial Intelligence for Automatic Defect Detection in Additively Manufactured Parts Using CT Scan Analysis.” *Journal of Intelligent Manufacturing*, vol. 36, no. 2, 2025, pp. 957–974. [DOI](#)

REFERENCES

Prof. Dr. Christian Huehne	Abteilungsleiter Funktionsleichtbau. Deutsches Zentrum für Luft- und Raumfahrt. Institut für Systemleichtbau
Prof. Dr. Oliver Voelkerink	Head of Institute., IMA, TU-Braunschweig
Dr. Nicola Cersullo	R&T Project Manager @ Airbus Hamburg